

# GUILHEM CARMOUZE

## Robotics engineering student, 3D vision and navigation

**SEEKING** 6-month end-of-studies internship, from March 2027. Robotics, 3D vision, autonomous navigation, robot perception, AI.

Toulouse, France  
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### EDUCATION

**Diplôme d'ingénieur, Robotic and Interactive Systems (SRI)** 2024 TO JUNE 2027  
UPSSITECH, engineering school of the University of Toulouse. French engineering degree, Master's level. Fifth and final year in 2026 to 2027.

**Preparatory cycle (CUPGE)** 2022 TO 2024  
UPSSITECH, University of Toulouse

### RESEARCH EXPERIENCE

**Research intern, Computer Vision Research Team** APRIL TO AUGUST 2026  
Artificial Intelligence Research Center (AIRC), AIST, Tsukuba, Japan. 29-page report in English: "Creation of a 360° Navigation Dataset Using 3D Gaussian Splatting".

- Built ArtiFixer-360, a pipeline from a plain pinhole video to a 360° video: COLMAP, a 3DGRUT Gaussian scene, a world-locked rig of 14 overlapping 110° views repaired jointly by a 14B video diffusion model, then distilled back into the scene, with reference-free quality gates. +23,602 lines and 119 tests over NVIDIA's upstream ArtiFixer; runs on 4-GPU nodes of the ABCI cluster.
- Depth-aware synchronisation lowered the cross-view depth-overlap error of the repaired views by 27% (0.0340 to 0.0247) before distillation. On a 117-frame reference run the temporal warp error (MAE, raw renders against output) fell from 0.037 to 0.020. The full 154-frame run failed my own acceptance gates and led to a geometry-first redesign.
- Earlier phases: A\* on a 5 cm occupancy grid and 4096 × 2048 panoramas rendered from six pinhole views, logged with position and heading, in a DISCOVERSE and MuJoCo simulation; a ROS 2 Humble interface prepared for a visual navigation model on the Kachaka robot (on main the model is not wired in; no run on the real robot is claimed).
- Also took part, with Alec Bossard, in extending SVLR (training-free visual language robotics, CNRS-AIST JRL) towards memory-dependent manipulation.

AI tools were used for research, code and spell-checking, as declared in the report; not for running experiments or producing results.

CODE [artifixer-360-pipeline](#) [nav\\_3dgs\\_pano](#) [KachakaNavigation](#)

### PROJECTS

**Usine 4.0, final-year team project** **IN PROGRESS** 2026 TO 2027  
Industry 4.0 smart factory (Usine 4.0), with my class at UPSSITECH.

**TLSe Racing, Formula Student driverless team** 2025 TO 2026

- My part: the simulation and tooling layer. A 2D Pygame simulator with a fit-to-track camera, a typed CSV cone-track loader and a configurable field-of-view sensor model that selects the visible cones.
- Also worked on cone-detection models (PyTorch) and on image-processing and control modules (ROS, Python, C++). The planners are my teammates' work: Alec Bossard and TJeanm.

CODE [TLSe\\_Racing\\_Driverless](#) [PathPlanning](#)

**Projet Fil Rouge, a real mobile robot** 2024 TO 2025  
Team of six: Arduino motor control, Raspberry Pi camera, LiDAR mapping with ICP, voice commands, ball tracking.

- My part: the web interface, a single-page app that drives the robot over the Web Bluetooth API; the Raspberry Pi MJPEG camera stream; the algorithm that turns the ball's image coordinates into drive commands to keep it centred; the timing of split voice commands.
- The semester before, with Alec Bossard: a colour-ball detector in pure C11, without OpenCV.

CODE [PFR2 \(team repository\)](#) [PFR](#)

### SKILLS

#### LANGUAGES

Python, C, C++, Java (coursework), JavaScript and TypeScript

#### LIBRARIES

PyTorch, NumPy, SciPy, OpenCV

#### 3D VISION

3D Gaussian Splatting (3DGRUT and 3DGUT, Splatfacto, DISCOVERSE), COLMAP, video diffusion pipelines (ArtiFixer), panoramic and equirectangular geometry

#### ROBOTICS

ROS 2 Humble, MuJoCo, path planning (A\*, occupancy grids), Kachaka API (gRPC), robot kinematics coursework (DH model and Jacobians of a 6-DOF UR3)

#### TOOLS

Docker, Singularity, PBS on an HPC cluster (ABCI), Git, Linux, pytest

### SIDE PROJECTS

Personal projects built in October 2026 with AI assistance; every number is reproduced by a script in the repository.

#### erpkIt

NumPy toolkit measuring what stitching views into a 360° panorama costs.

#### microslat

3D Gaussian Splatting in NumPy and PyTorch, one test per equation.

#### gaussian-projection-bench

EWA against unscented projection error, pinhole to equirectangular.

#### amr-traffic-lab

Robot traffic in an Industry 4.0 smart-factory (Usine 4.0) aisle.

### LANGUAGES

**French** native

**English** professional; 4-month research internship in Japan, in English

**Spanish** basic

### OTHER EXPERIENCE

#### Administrative assistant

2022 TO 2024

ALTINET, Tarbes